CIVE 536	Hydrology	3(3,0)
Measurement and analysis of precipitation data; flood analysis; methods for estimating evaporation and transpiration; streamflow measurement; groundwater hydrology; groundwater reservoirs; groundwater discharge; rainfall-runoff relationship; statistical analysis in hydrology. **Prerequisite**: CIVE 435. Annually.		

CIVE 538	Water and Wastewater Treatment	3(3,0)
Quality and control tests for water and wastewater, design principles of water and wastewater treatment plants, and physical, chemical, and biological operational processes. **Prerequisite**: CIVE 432, CIVE 435. Annually.		

CIVE 595A	BE Summative Learning Project 1	1(1,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisite**s: Senior Standing, ENGL 217. Annually.		

CIVE 595B	BE Summative Learning Project 2	3(3,0)
Team-oriented project that includes conducting and formulating experimental laboratory work and/or design problems on a specific, approved specialty topic. The project includes literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: CIVE 595A. Annually.		

II. Elective Courses

Students may satisfy their technical elective requirements by selecting from the following courses.

1. Common Courses and Construction Management Sequence

CIVE 502	Engineering Risk Analysis	3(3,0)
Elements of set theory, basic elements of probability theory, random variables, multiple random variables, functions of random variables, probabilistic models, statistical inference, and formulation of the reliability problem. **Prerequisite:** MATH 351.		

CIVE 503	Computer Methods in Civil Engineering	3(3,0)
Using the computer for analysis, design, and decision making in civil engineering. Conversion of CAD or REVIT elements to structural elements. Shop drawings. Value engineering. Applications. **Prerequisite:** Senior Standing.		

CIVE 504	Infrastructure Asset Management	3(3,0)
Problems of infrastructure, asset management principles, methods of inspection and monitoring of civil infrastructures, risk assessment, asset condition information and data, and the replace/maintain decision. **Prerequisite:** Senior Standing.		

CIVE 506	LEED and Green Construction Principles in Construction Management	3(3,0)
Green construction methods and benefits of applying the Leadership in Energy and Environmental Design (LEED) principles. **Prerequisite:** Senior Standing.		

CIVE 597	Topics in Civil Engineering	3(3,0)
This course covers topics of special interest in Civil Engineering. These topics may vary each time the course is offered, and the course content will be available with each course offering. May be repeated for credit as topics vary. **Prerequisite:** Consent of Advisor.		

2. Structural Analysis & Design Sequence

CIVE 510	Finite Elements Analysis	3(3,0)
Theory of the finite element method; spring, truss, and beam elements; constant-strain elements; plane-stress or plane-strain for axisymmetric problems; plates and shells; introduction to nonlinear material analysis; introduction to structural dynamic analysis; applications. **Prerequisite**: CIVE 312. On demand.		

CIVE 511	Structural Dynamics	3(3,0)
Dynamic response of single- and multi-degree-of-freedom systems; modal analysis of lumped and continuous mass systems; natural frequencies and mode shapes; deterministic time-domain approach and design response spectra; seismic response of structures; peak response; seismic design principles. **Prerequisite**: CIVE 312. On demand.		

CIVE 512	Advanced Structural Analysis	3(3,0)
Flexibility and Stiffness Matrix Methods for Analysis of Indeterminate Trusses, Beams, and Frames. Grid beams and structures on elastic supports. Composite Structural Systems. Influence lines for indeterminate structures using the concepts of virtual work and moment distribution methods – Qualitative and Quantitative approaches. **Prerequisite**: CIVE 507. On demand.		

CIVE 514	Earthquake Engineering	3(3,0)
Earthquake causes and measures; earthquake faults and waves; plate tectonics; structural dynamics of single- and multi-degree-of-freedom systems; seismic hazard evaluation in engineering practice; response analysis of structures. Equivalent static lateral force method, lateral load-resisting systems, architectural considerations for earthquakes, and mitigation of earthquake forces. **Prerequisite**: CIVE 312. On demand.		

CIVE 515	Prestressed Concrete	3(3,0)
Definitions and Methods of prestressing. Materials and their properties. Discussion of losses in prestressing. Elastic behavior of stress distribution under different load stages—analysis and design of homogeneous sections. Shear cracking load. Behavior		

of beam. Load Balancing Procedures. Deflection. Stresses resulting from shrinkage and creep. Partial prestressing. **Prerequisite**: CIVE 314. On demand.

CIVE 519	**Tall Building Structures**	**3(3,0)**
Loadings; structural systems and analysis modeling; braced frames; rigid frame structures; stability of high-rise buildings; dynamic response; shear wall structures; coupled shear wall structures; wall-frames; dual structural systems. **Prerequisite**: CIVE 413. On demand.		

3. Construction Material and Geotechnics Sequence

CIVE 521	**Special Concretes**	**3(3,0)**
Design and characteristics of lightweight concrete, heavyweight concrete, self-consolidating concrete, fiber-reinforced concrete, mass concrete; concrete in hot weather, concrete in cold weather, high-strength concrete, high-performance concrete, roller-compacted concrete, and shrinkage-compensating concrete. **Prerequisite**: CIVE 321. On demand.		

CIVE 522	**Durability of Construction Materials**	**3(3,0)**
Definition and importance of durability; properties of main building materials (concrete, ferrous and non-ferrous metals, wood, building stones, clay bricks, gypsum, lime, plastics); factors affecting durability; mechanisms of decay; precautions to increase durability; selection of durable material; durability tests; economic aspects and impact of durability. **Prerequisite**: CIVE 321. On demand.		

CIVE 523	**Repair of Concrete Structures**	**3(3,0)**
Conventional methods of investigation; nondestructive methods; structural health monitoring; corrosion risk evaluation; protection of reinforcing steel against corrosion; instrumentation techniques; repair materials and techniques; removal and preparation of concrete for repair; crack repair; concrete patching; structural strengthening; case studies. **Prerequisite**: CIVE 321. On demand.		

CIVE 524	**Nondestructive Evaluation & Instrumentation of Infrastructure**	**3(3,0)**
Overview of the different types of civil infrastructure, their problems and durability, use of nondestructive evaluation methods to assess their actual conditions, basic NDE methods, sonic and ultrasonic methods, ground penetrating radar (GPR), infrared thermography, electrical resistivity, radioactive and nuclear methods, sensing technologies, different instruments/sensors and their use, wireless technology, and structural health monitoring. Case studies and laboratory demonstrations. **Prerequisite**: CIVE 321. On demand.		

CIVE 525	**Concrete Technology**	**3(3,0)**
Cement and supplementary cementitious materials, mechanical behavior of concrete, properties and applications of various specialized concretes, hot weather concreting, cold weather concreting, chemical admixtures, concrete mix proportioning, deterioration mechanisms, environmental and physical causes of degradation, alkali-		

aggregate reaction, carbonation, chloride attack, sulfate attack, acid and seawater attack, freeze-thaw cycle. **Prerequisite**: CIVE 321. Annually.

CIVE 526	**Admixtures for Concrete**	**3(3,0)**
Definition and classification of concrete admixtures; precautions in their use; chemical, mineral, and miscellaneous admixtures; types and properties of admixtures; standard specifications; classification and purpose of use; beneficial and detrimental effects on the properties of fresh and hardened concrete; mechanisms of action; advantages and disadvantages**. Prerequisite**: CIVE 321. On demand.		

CIVE 527	**Seepage and Dams**	**3(3,0)**
Measurement of hydraulic conductivity (permeability), analysis of seepage using analytical solutions, graphical procedures, and numerical methods; control of seepage; wells, filters, and drains; types of dams; performance of earth and earth-rock dams. **Prerequisite**: CIVE 424; Co-requisite: CIVE 536. On demand.		

CIVE 528	**Earth Retaining Structures**	**3(3,0)**
Types of earth-retaining systems; earth pressure theory; design of externally stabilized walls; design of internally stabilized walls; reinforcing elements; fundamentals of soil-reinforcement interaction. **Prerequisite**: CIVE 424, CIVE 314. On demand.		

CIVE 529	**Advanced Foundation Engineering**	**3(3,0)**
Lateral earth pressure, retaining walls, sheet-pile walls, anchors, braced cuts, drilled-shaft foundations, foundations on collapsible soils, foundations on expansive soils, foundations on rock, soil improvement methods. **Prerequisite**: CIVE 424. On demand.		

4. Water and Environmental Engineering Sequence

CIVE 571	**Water Quality**	**3(3,0)**
Water quality parameters; sources of drinking water; drinking water problems and their impact on human life; drinking water treatment methods; common operations in water treatment plants; design of distribution systems; assessment and monitoring of water quality. **Prerequisite**: CIVE 432. On demand.		

CIVE 573	**Solid Waste Management**	**3(3,0)**
Study of the types and sources of non-hazardous municipal solid waste; estimation of waste quantities and storage volumes; identification of collection, transfer, treatment, and disposal alternatives; preliminary design of landfills; solid waste management principles and processes; and pollution control management. **Prerequisite**: CIVE 432. On demand.		

CIVE 575	**Water Supply & Wastewater Systems**	**3(3,0)**
Identification and evaluation of water supply solutions; design of water distribution networks; pumping; population forecasting and requirements; collecting and		

disposing of stormwater and sewage; design of storm and sewage systems; excess stormwater management; environmental risk and protection measures. **Prerequisite**: CIVE 435. On demand.

CIVE 576	**Hydraulic Structures**	**3(3,0)**
Hydraulic aspects of the theory and design of hydraulic structures; types of dams and their functions; stability analysis of gravity dams; culverts, navigation canals, reservoirs, and port facilities; and control structures such as gates and weirs. **Prerequisite**: CIVE 435 and CIVE 536. On demand.		

5. Public Works Sequence

CIVE 540	**GIS for Civil Engineering**	**3(3,0)**
Review of basic cartographic principles and the use of geographic information systems for thematic mapping, spatial analysis, and application in the water resources sector. The laboratory emphasizes experience with GIS software. **Prerequisite**: Senior Standing. On demand.		

CIVE 541	**Pavement Design**	**3(3,0)**
Analysis and design of flexible and rigid pavements; pavement type selection; loading; failure criteria and reliability; mechanistic pavement design; design exercises using existing methods. **Prerequisite**: CIVE 423. On demand.		